



The role of the intestinal microbiota in allogeneic HCT: clinical associations and preclinical mechanisms

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Allogeneic hematopoietic cell transplantation (allo-HCT) is a curative-intent therapy for patients with hematological malignancies, but despite advances in the field in recent years, there is still a significant risk of post-transplant mortality. In addition to relapse of the underlying malignancy, the key contributors to this high mortality are graft-versus-host disease (GVHD) and infection. The intestinal microbiota is the collective term describing the community of bacteria, fungi, viruses and protozoa that resides in the human gastrointestinal tract. Bacterial communities have been studied most comprehensively, and disruption of these communities has been associated with the development of a variety of medical conditions in large clinical associative studies. Preclinical studies suggest a mechanistic role for the intestinal microbiota in the instruction and maintenance of both intestinal and systemic immune cell function. This review outlines our current understanding of the relationship between gut bacteria and allo-HCT outcomes, including infection, immune reconstitution, GVHD and relapse, drawing on evidence from both clinical associative studies and preclinical mechanistic studies.

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their host, due to direct contact with both parenchymal and hematopoietic cell populations, and their capacity to produce circulating metabolites. Advances in sequencing technologies in recent years have facilitated the rapid progress of clinical associative studies and these, in turn, have served as valuable hypothesis-generating tools for mechanistic studies in preclinical animal models.

Allogeneic hematopoietic cell transplantation (allo-HCT) is a curative-intent treatment for patients with hematological malignancies. However, it is a high-risk therapy, as overall post-transplant mortality remains in the order of 50%, most commonly due to disease progression or relapse, infection and graft-versus-host disease (GVHD) [1]. Microbial diversity, a summary measurement of the richness and evenness of unique bacterial taxa, is a useful measure of intestinal microbiota health. Several groups have observed a decline in intestinal microbial diversity manifesting early in the course of allo-HCT, and this injury to the intestinal microbiota is associated with lower overall survival after allo-HCT [2,3^{••}]. Exposures to broad-spectrum antibiotics are associated with a decrease in the intestinal microbiota biodiversity during allo-HCT, although other factors such as dietary changes, conditioning regimens and other medications are likely to influence the post-HCT intestinal microbiota dynamics (Table 1) [4,5].

One of the first indications that the intestinal microbiota could play a role in transplant outcome came from the observation that transplanted germ-free mice had improved survival compared to specific-pathogen free mice with intact microbiota [6]. Since then, clinical strategies aimed at ‘decontaminating’ the gut have been an area of active investigation, although these clinical trials have produced conflicting results, suggesting the need for mechanistic studies using preclinical models to fully identify important taxonomic groups and their interactions with the human host during allo-HCT [7]. Several clinical studies have reported associations between specific features of the intestinal microbiota and transplant outcomes, some favorable and some unfavorable [3^{••},8–13]. Many currently active clinical trials employ therapeutic strategies that aim to eliminate microbiome features associated with adverse transplant outcomes, while preserving features associated with desirable outcomes, as summarized in Table 2.

This review outlines our current understanding of the role of the intestinal microbiota in allo-HCT, focusing on four important clinical outcomes: infection, immune

Introduction

The digestive tract harbors the largest microbial community of the mammalian body, and these micro-organisms have significant capacity to interact with and influence

Table 1

Potential modulators of the intestinal microbiota during allo-HCT

Factors	Impacts	References
Antibiotics		
Broad-spectrum antibiotics (β -lactam, metronidazole, meropenem)	Broad-spectrum antibiotics target obligate anaerobes which are typically producers of short-chain fatty acids in the gut, allowing the expansion of opportunistic pathogens	[4,55]
Imipenem-cilastatin	Imipenem-cilastatin treatment in mice leads to expansion of the mucus-degrading bacteria <i>Akkermansia</i>	[55]
Fluoroquinolones	Prophylactic fluoroquinolones lead to a decreased risk of intestinal domination by Gram-negative bacteria	[10]
Transplant-associated factors		
Conditioning chemotherapy: carmustine, etoposide, aracytin and melphalan	After conditioning, the intestinal microbiota diversity declined, accompanied by a decrease in <i>Faecalibacterium</i> abundance and an increase in <i>Escherichia</i> abundance	[72]
Conditioning intensity	Patients receiving non-myeloablative conditioning regimens have higher intestinal diversity at engraftment time compared to patients receiving myeloablative conditioning regimens	[5]
Total body radiation	Total body radiation in mice leads to reduction in the intestinal microbiota diversity and a shift in the overall microbial compositions compared to controls	[73]
Immuno-suppressive medications (methotrexate, cyclosporine A)	In a high-throughput <i>in vitro</i> drug screen, some human-target drugs show antibacterial activities to some commensal bacterial strains often found in the human gut	[74]
Diet		
Total parenteral nutrition (TPN)	Use of long-term TPN (≥ 10 days) is associated with a loss of <i>Blautia</i>	[8]
Western diet/fiber-free diet	Mice fed on a prolonged fiber-free diet experience decreased intestinal microbiota diversity, a decrease in <i>Bifidobacteria</i> and an increase in mucus-degrading <i>Akkermansia</i>	[30,31]
Dietary fiber inulin supplementation	Inulin supplementation restores the mucus layer in mice fed on a Western diet	[75]
High-fiber (pectin, cellulose) diet	High-fiber diets containing cellulose and pectin lead to increased cecal and systemic levels of short-chain fatty acids such as acetate, propionate and butyrate	[76]
Host genetics		
rs4988235 SNP conferring lactase expression and lactose intolerance in European populations	C/C SNP is associated with non-absorbers of lactose and increased risk of <i>Enterococcus</i> domination after allo-HCT	[54**]
Paneth cell α -defensin-5 gene SNPs	rs4415345G is associated with a higher abundance of a butyrogenic obligate anaerobe, <i>Odoribacter splanchnicus</i> , which is associated with a reduced risk of acute GVHD	[77*]
NLRP6 deficiency	NLRP6 deficiency in the intestinal epithelial cells leads downregulation of IL-18 and alters the AMP production to a profile that favors intestinal microbiota dysbiosis	[33]

reconstitution, GVHD and relapse (Figure 1). Allo-HCT patients are at an increased risk of infection due to high degree of immunosuppression and prolonged hospital admissions, and several studies have suggested that the intestinal commensals play a role in preventing intestinal domination by both opportunistic pathogens and commensal bacteria, which can lead to subsequent bloodstream infections [14–16]. Acute GVHD (aGVHD) is a systemic, potentially fatal condition that occurs following allo-HCT, in which donor-derived T cells recognize host antigens as foreign, resulting in cytotoxic tissue damage to predominantly the skin, the liver and the gastrointestinal tract [17]. Reconstitution of the immune system is critical for the recovery of transplant patients, and aberrant immune reconstitution can lead to increased late infections, increased GVHD incidence and failure of the graft-versus-tumor

(GVT) effect [18]. Lastly, the GVT effect following allo-HCT, in which graft-derived T cells and NK cells eliminate host malignant cells, is critical in ensuring long-term survival in allo-HCT patients.

How does the intestinal microbiome interact with the mammalian host?

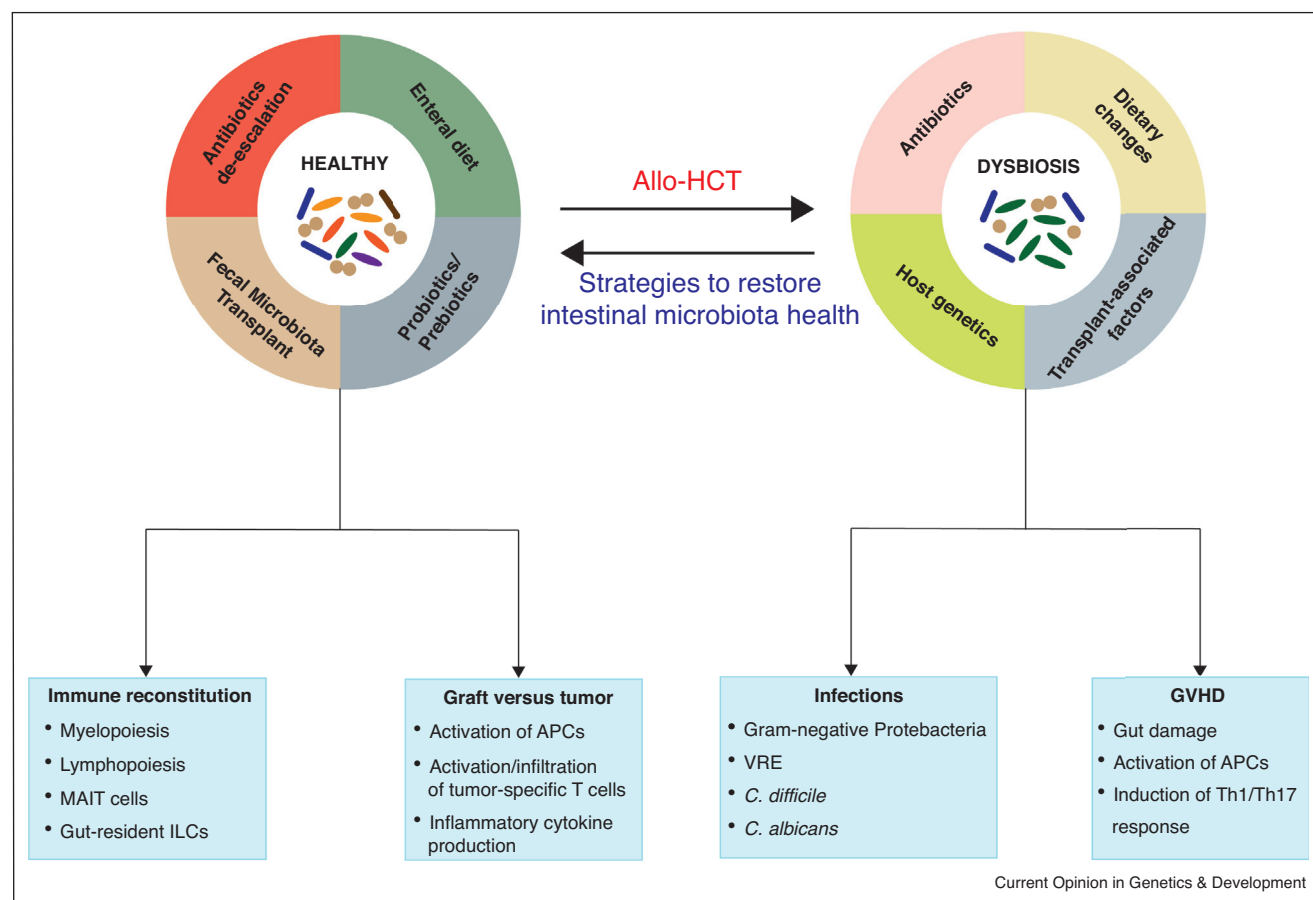
The intestinal bacterial community consists of over 2000 species, the majority of which belong to phyla Proteobacteria, Firmicutes, Actinobacteria and Bacteroidetes [19,20]. Gut bacterial communities are highly variable in healthy people, but are thought to remain relatively stable within individuals over time. In healthy adults, the inter-personal variation in the gut bacterial community structure can be partially, though not entirely, explained by differences in diet [21,22,23*].

Table 2

Intestinal microbiota-focused clinical trials in allo-HCT patients

Trial title	Phase	Aims and intervention	NCT number/references
Infection			
Autologous fecal microbiota transplantation (Auto-FMT) for prophylaxis of <i>Clostridium difficile</i> infection in recipients of allogeneic hematopoietic stem cell transplantation	II	Assess the efficacy of auto-FMT for prevention of <i>C. difficile</i> infection in allo-HCT patients	NCT02269150 [37]
Oral supplementation of 2'-fucosyllactose in allogeneic bone marrow transplant recipients	I/II	Assess whether 2'-fucosyllactose supplementation lead to lower incidence of acute GVHD and blood-stream infection at day 100 post-allo-HCT	NCT04263597
Oligosaccharide for Cdiff(+) heme-onc patients		Assess the feasibility of oligosaccharide supplementation (potato starch) in preventing <i>C. difficile</i> colonization	NCT03778606
GVHD			
<i>Lactobacillus rhamnosus</i> GG in reducing incidence of graft-versus-host disease in patients who have undergone donor stem cell transplant		Assess the treatment of allo-HCT with a probiotic <i>Lactobacillus rhamnosus</i> GG in reducing incidence of lower GI acute GVHD	NCT02144701 [78]
Dietary manipulation of the microbiome-metabolomic axis for mitigating GVHD in allo HCT patients	II	Evaluate the feasibility, safety and efficacy of administering potato-based resistant starch to allo-HCT patients	NCT02763033
Study of IL-22 IgG2-Fc (F-652) for subjects with grade II-IV lower GI aGVHD	I/II	Investigate the safety and pharmacokinetics of F-652 (in combination with corticosteroids) in allo-HCT patients with grade II-IV lower GI aGVHD	NCT02406651
Prebiotic galacto-oligosaccharide and acute GVHD		Determine whether the prebiotic galacto-oligosaccharide can modulate the intestinal microbiota and help prevent GVHD	NCT04373057
Lipopolysaccharide metabolism and identification of potential biomarkers predictive of graft-versus-host disease after allogeneic stem cell transplantation (MetAlloLip)		Assess whether LPS activity index can be quantified early after allo-HCT and predictive of the risk of severe GVHD	NCT03918343
<i>Lactobacillus plantarum</i> in preventing acute graft-versus-host disease in children undergoing donor stem cell transplant	III	Assess the efficacy of orally administered <i>Lactobacillus plantarum</i> in preventing the development of GI acute GVHD	NCT03057054
Fructooligosaccharides in treating patients with blood cancer undergoing donor stem cell transplant	I	Evaluate the side effects and optimal dose of prebiotic fructooligosaccharides in the prevention of acute GVHD	NCT02805075
Individualized nutrition for adult recipients of allogeneic stem cell transplantations - effects on quality of life (NASQ)		Assess the effect of lactose-reduced and energy-enriched oral nutrition on acute GVHD incidence, engraftment time and neutropenic fever	NCT01181076
Human lysozyme goat milk for the prevention of graft versus host disease in patients with blood cancer undergoing a donor stem cell transplant		Assess the feasibility, toxicity and efficacy of human lysozyme goat milk in the prevention of GVHD	NCT04177004
Immune reconstitution			
Dysbiosis and immune reconstitution after allo-HSCT (PARI-DYS)		Explore the changes in the intestinal microbiota and early post-transplant reconstitution of invariant NKT cells	NCT03616015
Monitoring of immune and microbial reconstitution in (HCT) and novel immunotherapies		Correlate microbiota changes and their interactions with immune reconstitution and immune functions after allo-HCT	NCT03557749
Strategies to manipulate the gut microbiota during allo-HCT			
Aims and intervention		NCT number/references	
Fecal microbiota transplantation in restoring the intestinal microbiota health and in preventing infections and acute high-risk GVHD during allo-HCT		NCT03720392; NCT04014413; NCT03359980; NCT03678493; NCT04139577; NCT03549676; NCT04059757; NCT03812705; NCT04285424; NCT04280471; NCT03819803; NCT04269850	
Optimizing antibiotics treatment to minimize intestinal microbiota damage, prevent subsequent infections and GVHD during allo-HCT		NCT03733340; NCT03078010; NCT03727113; NCT02641236	

Figure 1



Intestinal microbiota dysbiosis and patient outcomes following allo-HCT.

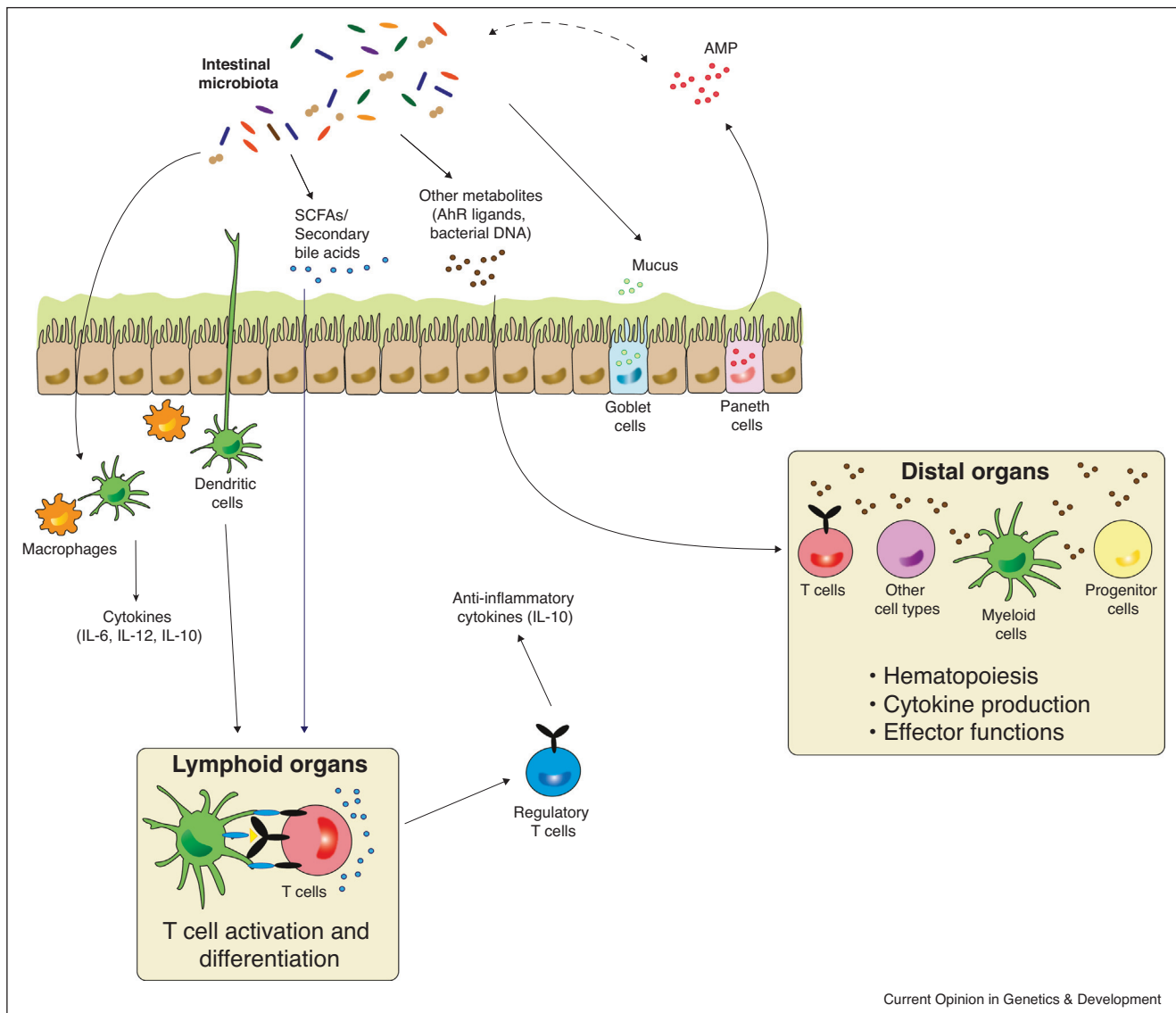
Allo-HCT patients are exposed to various environmental conditions including cytotoxic conditioning regimens, antibiotics, and dietary changes that might contribute to changes in the intestinal microbiota. These injuries to the intestinal microbiota, in turn, are associated with transplant outcomes such as infections, immune reconstitution, GVHD and relapse through various different immunological mechanisms involving different hematopoietic and non-hematopoietic cell populations. Strategies to restore the intestinal microbiota health include fecal microbiota transplantation, de-escalated antibiotic exposures and enteral diets that promote the growth of beneficial bacteria. VRE, vancomycin-resistant *Enterococcus*; MAIT, mucosal-associated invariant T cell; ILC, innate lymphoid cell; APC, antigen-presenting cell; Th, T helper.

The majority of studies of the role of the intestinal microbiota in allo-HCT outcome have utilized 16S rRNA gene amplicon sequencing, which relies on pre-amplification of the hypervariable regions of the 16S rRNA gene before sequencing. This technique allows identification of the intestinal microbiota at genus-level resolution, and is practical to scale to a large collection of fecal samples. The disadvantage of 16S amplicon sequencing, however, is the lack of taxonomic resolution at species and strain-specific levels [24]. To address this challenge, whole-genome metagenomic sequencing can be performed, which provides the genomes of all micro-organisms present in fecal samples, resulting in high-resolution profiles of the intestinal bacterial compositions and predictions of functional potential of these micro-organisms [25–27]. In addition, advances in metabolomics have allowed further

assessment of the function of intestinal bacteria by measuring actual metabolites produced by the gut microbes. Together, sequencing and metabolomic analyses of clinical samples can generate high-quality hypotheses to be tested in pre-clinical mechanistic models of human health and disease [20,28,29].

The gut commensals can influence local and systemic immunity through both direct interactions with immune cells, and indirect interactions mediated via secreted microbial metabolites, such as short-chain fatty acids (SCFAs), secondary bile acids and other small molecules (summarized in Table 1 and Figure 2) [30]. Short-chain fatty acids (SCFAs), such as butyrate, propionate and acetate, are produced as a result of the fermentation of dietary fiber by the intestinal microbiota. Butyrate (and to

Figure 2



Interactions between the intestinal microbiota and the mammalian host.

Within the gut lumen and intestinal epithelial environment, the intestinal microbiota promotes mucus production from goblet cells, as well as antimicrobial peptides production from Paneth cells to maintain gut homeostasis. Antimicrobial peptides, in turn, regulate the intestinal microbial compositions. Signals from the intestinal microbiota also activate antigen-presenting cells such as macrophages and dendritic cells, leading to downstream production of cytokines that promote context-specific inflammation or tolerance. Dendritic cells can also migrate to lymphoid organs to present antigens and activate T cells, and microbially derived metabolites such as SCFAs and secondary bile acids could mediate this process by influencing T cell activation and differentiation. Other microbially derived products such as AhR ligands, bacterial DNA and bacterial components can also translocate across the intestinal epithelial barrier into systemic circulation, reaching distal organs such as the liver, the lung and the bone marrow and modulate the generation, activation and differentiation of different cell subsets at these sites. SCFA, short-chain fatty acid; AhR, aryl-hydrocarbon receptor; AMP, anti-microbial peptide; IL, interleukin.

a lesser extent propionate) acts as a histone deacetylase inhibitor to promote the differentiation of colonic FoxP3⁺ regulatory T cells (T_{reg}) in mice [31,32]. Propionate and acetate also induce anti-inflammatory responses and improve intestinal epithelial barrier integrity [33,34]. Secondary bile acids, which are produced as a result of

microbial metabolism of primary bile acids excreted by the liver, can also induce differentiation of peripheral T_{regs} [35,36]. Microbe-derived indole derivatives from tryptophan metabolism confer a protective effect against epithelial barrier damage after total body irradiation (TBI) and in a chemically induced colitis model

[37,38]. Moreover, the intestinal microbiota also produces many metabolites that act as ligands for the aryl hydrocarbon receptors (AhR), which are expressed by many cell types, and can play an important role in the induction and development of immune-mediated diseases (Figure 2) [39–41].

The intestinal microbiota contributes to the maintenance and degradation of the mucus layer in the GI tract, which forms a protective barrier between microbes and intestinal epithelial cells (IECs). However, the outgrowth of bacteria with mucus-degrading activities can potentially lead to impaired intestinal barrier integrity [42–44]. In addition, the intestinal microbiota also modulates antimicrobial peptide (AMP) production by intestinal epithelial cells (IECs), such that disruption of the indigenous microbiota can affect AMP expression to favor the expansion of pathogenic bacteria in mice (Figure 2) [45]. Alternatively, host genetics can also modulate the intestinal microbiota compositions: NLRP6 deficiency in mice leads to aberrant AMP production that favors intestinal microbiota dysbiosis (Table 1) [45]. Gut commensals can also directly induce the expression of critical innate immune effectors and AMPs to maintain intestinal mucosal integrity and prevent colonization by opportunistic pathogens [46,47]. The intestinal commensals, by themselves, also confer colonization resistance against opportunistic pathogens by nutrient limitation and bacteriocins [14,48,49].

The intestinal microbiota and infections after allo-HCT

Clinical evidence

Infections account for about 30% of deaths within the first 100 days after allo-HCT [1]. During allo-HCT, intestinal microbiota diversity rapidly declines, commonly accompanied by domination of a single bacterial taxon such as members of the *Enterococcus* and *Streptococcus* genera, and members of the class Proteobacteria [5,10]. Domination of the intestinal flora by organisms in the class Proteobacteria, as well as by organisms in the genus *Enterococcus* such as vancomycin-resistant *Enterococcus* (VRE), commonly precedes, and is predictive of, subsequent bacterial blood-stream infections by Gram-negative bacteria and VRE bacteremia, respectively [4,50]. Exposure to fluoroquinolone prophylaxis was associated with a decreased risk of intestinal domination by Proteobacteria, as well as a decreased risk of Gram-negative infection in a multivariate analysis controlled for graft source, underlying disease and other antibiotic prophylactic regimens (Table 1) [4,10].

Given the now well-established decline in microbiota diversity following allo-HCT and its association with poor transplant outcomes, fecal microbiota transplantation (FMT) represents a potentially efficacious intervention to restore the intestinal microbiota health following allo-

HCT [3^{••},5,11,12,51^{••}]. While the efficacy and safety of FMT remains under investigation, this procedure has proved to be effective to treat recurrent *Clostridium difficile* infection (CDI) [51^{••},52]. Higher relative abundance of three specific bacterial families, *Lachnospiraceae*, *Ruminococcaceae* and *Bacteroidaceae*, is associated with a reduced risk of CDI in allo-HCT patients [53]. These bacteria are some of the most abundant families in the intestinal microbiota of healthy people.

A recent clinical study described an association between the intestinal microbiota and fungal infection with *Candida* spp. during allo-HCT. Specifically, a decrease of total bacterial biomass and biodiversity, as well as loss of *Lachnospiraceae* and *Bacteroidetes*, is associated with increased relative abundance of *Candida* spp [54^{••}]. Increased intestinal abundance of butyrate producers, such as *Clostridiales* and *Faecalibacterium*, has been associated with a decreased risk of lower respiratory tract infection in allo-HCT patients [55].

Preclinical evidence

There are few preclinical studies focusing on the mechanisms by which the intestinal microbiota influences infections following allo-HCT. Microbe-derived metabolites and small molecules such as SCFAs, secondary bile acids and bactericidal toxins can directly inhibit the growth and colonization of opportunistic pathogens such *C. albicans* and VRE [14,46,56,57]. However, it is necessary to validate these findings in the specific context of allo-HCT in order to develop appropriate microbiome-targeted interventions.

Intestinal microbiota and immune reconstitution after allo-HCT

Clinical evidence

The influence of the intestinal microbiota on immune reconstitution in allo-HCT patients is an important and active area of research in the transplant field; however, thus far there are limited clinical data available. In a high-resolution computational analysis combining longitudinal data of total white blood cell count and fecal 16S sequencing from 2047 allo-HCT patients, *Bacteroidaceae*, *Prevotellaceae* and *Ruminococcaceae* were associated with increased lymphocyte, monocyte and neutrophil counts, respectively [58^{••}]. Furthermore, high intestinal microbial biodiversity, high abundance of *Blautia* and *Bacteroidetes*, as well as high relative abundance of genes associated with the riboflavin synthesis pathway, have been associated with increased numbers of mucosal-associated variant T (MAIT) cells following allo-HCT in patients, although these findings need to be interpreted with caution due to small sample sizes [59,60[•]].

Preclinical evidence

Depletion of the intestinal microbiota with broad-spectrum antibiotics leads to impaired recovery of lymphocytes and

neutrophils after total body irradiation (TBI) in mice, an effect which was able to be restored with caloric supplementation. Specifically, treatment of transplanted mice with sucrose supplementation did not significantly alter the total fecal bacterial abundance or the compositions of the gut flora, suggesting that the intestinal microbiota potentially provides additional nutrition from the diet to support early immune reconstitution [61]. At steady state, antibiotic-treated mice show significant bone marrow suppression and reduced peripheral lymphocyte counts, potentially an effect mediated via STAT1 signaling in progenitor cells [62], however, these findings have yet to be examined in the setting of allo-HCT.

Intestinal microbiota and graft-versus-host disease (GVHD)

Clinical evidence

Several studies have reported that increased microbial diversity during the peri-neutrophil engraftment period is associated with a decreased risk of GVHD-related mortality [3[•],5,11–13]. The intestinal microbial diversity during allo-HCT is most commonly determined by analyzing 16S sequencing data from fecal samples and is a summary measure that reflects both the number of unique bacterial taxa and their relative abundance. Indoxyl sulfate, a product of the tryptophan metabolism by the intestinal microbiota, is a potential biomarker of intestinal biodiversity which can be measured in urine and which has been studied in a pilot study of 31 allo-HCT patients [11]. Decreased concentrations of these indole-derived metabolites were observed in allo-HCT patients with active aGVHD compared to patients without aGVHD [63^{••}].

Several specific members of the intestinal microbial community have been associated with either increased or decreased rates of GVHD. SCFA-producing members of the families *Lachnospiraceae* and *Ruminococcaceae*, such as *Blautia*, *Clostridium* and *Lachnospiraceae*, are associated with protection against aGVHD-related mortality and chronic GVHD development [8,64]. The absence of these beneficial bacteria attributed to broad-spectrum antibiotics is correlated with an increased risk of GVHD development, specifically in the GI tract [65,66]. High relative abundance of *Lachnospiraceae* and *Ruminococcaceae* has also been correlated with an increased ratio of T_{reg}/Th17 cells, potentially maintaining the balance of anti-inflammatory and pro-inflammatory responses during GVHD [65,67].

In addition, some bacterial taxa have been associated with increased GVHD severity. *Enterococcus* domination post-allo-HCT is associated with an increased risk of severe GVHD and GVHD-related mortality in allo-HCT patients, as well as in mice [11,68^{••}]. Furthermore, *Enterococcus* requires lactose as a source of nutrition for maximal growth, and patients with the single-nucleotide

polymorphism associated with lactose intolerance (thus retention of lactose in the gut lumen) also saw an increased risk of *Enterococcus* domination following allo-HCT [68^{••}]. *Akkermansia* expansion, which is associated with exposure to piperacillin/tazobactam and imipenem during transplant, is also correlated with an increased risk of GVHD-related mortality in both humans and mice [42,69]. Moreover, this genus has been associated with an increased risk of chronic GVHD [64].

Preclinical evidence

The intestinal microbiota potentially makes a mechanistic contribution to the development of GVHD through multiple mechanisms. Tissue damage caused by conditioning regimens allow translocation of pathogen-associated molecular patterns such as lipopolysaccharide (LPS) across the intestinal barrier, which can activate the innate immune system and is thought to be key for initiation of GVHD [70,71]. Exposure to imipenem leads to decreased gut barrier integrity, resulting in more severe GVHD in mice, potentially due to the resultant increase in relative abundance of *Akkermansia* [69]. Treatment with indole-3-carboxaldehyde, an indole derivative, limits gut epithelial damage and prevents the initial cascade of inflammatory responses triggered by TBI in a mouse model of GVHD [37]. Other indole derivatives can also activate AhR on innate lymphoid cells (ILCs) and induce IL-22 expression, which promotes epithelial regeneration after radiation and improves GVHD survival in mice [39,40,72].

In addition, propionate, and to a lesser extent butyrate, has been shown to promote the activation of the NLRP3 inflammasome to support intestinal crypt regeneration post-allo-HCT [73]. Oral dosing of mice with 17 butyrate-producing *Clostridium* strains after allo-HCT restores the intestinal butyrate levels and improves intestinal epithelial integrity, thereby reducing GVHD severity [74]. Oral administration of butyrate in a mouse model of allo-HCT also promotes epithelial cell regeneration and mitigates GVHD [74].

Bacteria signal through toll-like receptors to recruit neutrophils into the ileum after allo-HCT, where they generate reactive oxygen species that further amplify radiation-induced damages in the GI tract of recipient mice [75]. Moreover, neutrophil migration from the ileum to the mesenteric lymph nodes also depends on the intestinal microbiota, promoting alloantigen presentation to donor T cells [76]. The intestinal microbiota also promotes MHC class II expression on IECs in a MyD88/TRIF-dependent manner, following the cues of inflammatory cytokines IFN- γ and IL-12, thereby increasing alloantigen presentation to donor T cells [77[•]]. Furthermore, professional donor-derived APCs such as CD103⁺ DCs migrate from the lamina propria to the mesenteric lymph nodes, where they promote the activation and proliferation of donor T cells through antigen presentation,

and release inflammatory cytokines IL-6 and IL-12 in a MyD88/TRIF-dependent manner, suggesting the involvement of the intestinal microbiota in the activation and migration of APCs in GVHD pathophysiology [78]. Pre-transplant *Enterococcus* colonization led to increased GVHD severity and mortality in a germ-free mouse model of GVHD and furthermore, feeding experimental mice a lactose-free diet prevented *Enterococcus* growth while preserving other members of the gut commensals, leading to improvement in GVHD-related survival in a mouse model of allo-HCT [68^{**}]. The specific mechanism by which *Enterococcus* modulates post-transplant immune function and influences alloreactive T cells during GVHD remains unclear and requires further investigation [68^{**}].

Not unsurprisingly, the influence of the complex bacterial community in the GI tract on GVHD outcome is complex, with some bacterial taxa appearing to be protective and others pathogenic. This highlights the need for novel therapeutic approaches which enhance the protective taxa while eliminating or decreasing the pathogenic taxa.

Intestinal microbiota and graft-versus-tumor (GVT) effect

Clinical evidence

Relapse and/or progression of disease remains the most common cause of death in allo-HCT patients, accounting for 51–59% of mortality at or after day 100 post-transplant [1]. Features of the intestinal microbiota might serve as a potential biomarker of relapse following allo-HCT [31,35,79,80]. In the context of allo-HCT, a cluster of bacteria including *Eubacterium limosum* has been associated with a decreased risk of relapse in a multivariate analysis of 541 patients from MSKCC, where the analysis was controlled for graft source, conditioning intensity, HLA-match and disease [9]. *E. limosum* is a producer of butyrate, propionate, acetate and lactate, and has previously been demonstrated to attenuate the severity of chemically induced colitis [81]. Whether this particular species may influence GVT responses after allo-HCT remains unclear and should be explored further.

Preclinical evidence

Several studies using mouse models have suggested that the intestinal microbiota might influence anti-tumor activities of immunotherapies outside the context of allo-HCT [82^{*},83,84,85^{*}]. It is necessary to follow up on the finding regarding the influence of *E. limosum* on the graft-versus-tumor effects with a mechanistic study using mouse models of allo-HCT, as well as exploring the role of other bacterial taxa in the anti-tumor activities in the setting of allo-HCT.

Therapeutic interventions

There are currently a number of clinical trials underway that target the intestinal microbiota in the allo-HCT patient population (summarized in Table 1). Multiple

clinical trials are exploring the safety and efficacy of pre-transplant and post-transplant FMT in the prevention of *C. difficile* and other infections in allo-HCT patients, as well as exploring microbiome-targeted strategies for prevention and treatment of steroid-refractory acute GVHD. Optimization of antibiotic regimens to limit the damage to the gut flora is also being actively investigated in the clinic. While previous studies indicate that the intestinal microbiota states during the pre-transplant and peri-neutrophil-engraftment periods can predict patient outcomes, the timeframe in which microbiome-targeted therapies may provide maximal benefits requires further investigations in clinical trials [3^{**}]. Additionally, multi-omic approaches, incorporating metagenomics and metabolomics in the context of well-designed clinical trials will further expound on the potential mechanisms by which the intestinal microbiota influences allo-HCT outcomes.

Conclusion

Clinical investigations have described relationships between the intestinal microbiota during allo-HCT and multiple patient outcomes, most notably GVHD and infection, but also immune reconstitution and GVT. Preclinical studies have furthered these clinical findings by elucidating some key mechanisms, most importantly the activation and increased antigen-presenting functions of APCs, effector functions of T cells and key mediated cytokines such as IL-6 and IL-12. However, knowledge gaps persist that could potentially limit the clinical application of microbiota-modulating therapies. Specifically, it is unclear whether the intestinal microbiota needs direct contact with immune cells to exert effects, or whether these interactions are mediated through microbially derived metabolites which can be potentially neutralized with inhibitors where they appear pathogenic, or directly administered where their function is protective. Furthermore, the critical time points in which the intestinal microbiota exerts its influence on outcomes in allo-HCT remain unclear, and this is critical data that is required to determine the time window for clinical interventions. The future of microbiota research in the context of allo-HCT will combine the insights of clinical and preclinical studies, and translating these observational and mechanistic understandings into clinical trials will be key to developing microbiome-targeted strategies to maximize patient outcomes following allo-HCT.

Conflict of interest statement

Dr van den Brink reports grants from NCI: P01 CA023766-40, grants from NCI: P30 CA008748-54, grants from NCI: R01 CA228308-03, grants from NIAID: U01 AI124275-05, grants from Parker Institute for Cancer Immunotherapy, Co-Director, grants, personal fees and other from Seres Therapeutics, during the conduct of the study; personal fees from Merck and Co, Inc., personal fees from Acute Leukemia Forum, personal fees from Jazz Pharmaceuticals, personal fees from Therakos,

personal fees from Amgen, personal fees from PureTech, personal fees from Straximm, personal fees from Rubius Therapeutics, personal fees and other from DKMS, personal fees from Magenta Therapeutics, personal fees from WindMIL Therapeutics, grants from NIA: P01 AG052359-04, grants from NHLBI: R01 HL123340-06, grants from NHLBI: R01 HL125571-05, grants from NCI: R01 CA228358-03, grants from Tri-Institutional Stem Cell Initiative award 2016-13, grants from The Lymphoma Foundation, grants from The Susan and Peter Solomon Divisional Genomics Program, personal fees from Mallinckrodt Pharmaceuticals, personal fees from Kite Pharma, Inc., grants from NHLBI: R01 HL147584-02, grants from Cycle for Survival, grants from Starr Cancer Consortium, grants from Hyundai Hope on Wheels, grants from Tri-Institutional Stem Cell Initiative, outside the submitted work. In addition, Dr van den Brink has a patent Intestinal *Blautia* and reduced GVHD-related Mortality with royalties paid to Seres Therapeutics, a patent Methods and Compositions for Detecting Risk of Cancer Relapse with royalties paid to Seres Therapeutics, and a patent Bacterial Compositions and Methods for Cancer Survival pending. Dr van den Brink has stock options with Seres Therapeutics.

Declaration of Competing Interest

The authors report no declarations of interest.

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- of special interest
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